

UrbanSpire AI

Elevating city operations
through intelligence.

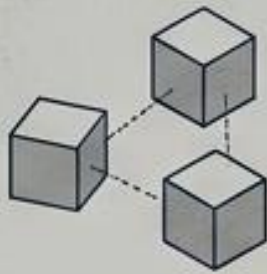
Capstone Project
Lenovo LEAP NextGen Scholar Program

Presented By
Radhika Mahajan

Individual Capstone Project

The Friction in Urban Governance

Urban local bodies currently operate in silos, creating a cascade of systemic challenges:



Fragmented Systems

Multiple disconnected departmental interactions.



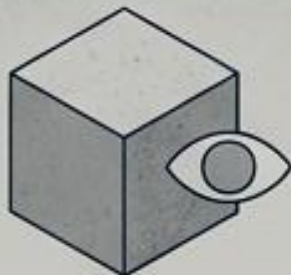
Slow Resolution

Significant delays in complaint registration and processing.



Manual Inefficiency

Paper-heavy processing of certificates and core public services.



Opacity

A fundamental lack of transparency in service delivery.



The AI Gap

Limited to no utilization of Artificial Intelligence for citizen support.



Blind Spots

Severe difficulty in monitoring service performance across departments.

A Unified Digital Ecosystem

An integrated, AI-powered municipal governance platform.

UrbanSpire AI pulls fragmented civic services into a single, intelligent hub—simplifying the citizen experience while supercharging operational efficiency for government departments.



For Citizens:

A seamless digital front door to the city.



For Government:

A centralized command center for urban management.

The Core Pillars of UrbanSpire AI

Citizen Services

- ☑ - Birth Certificate Applications
- ☑ - Property Tax Management
- ☑ - Online Complaint Registration & Tracking

AI & Communication

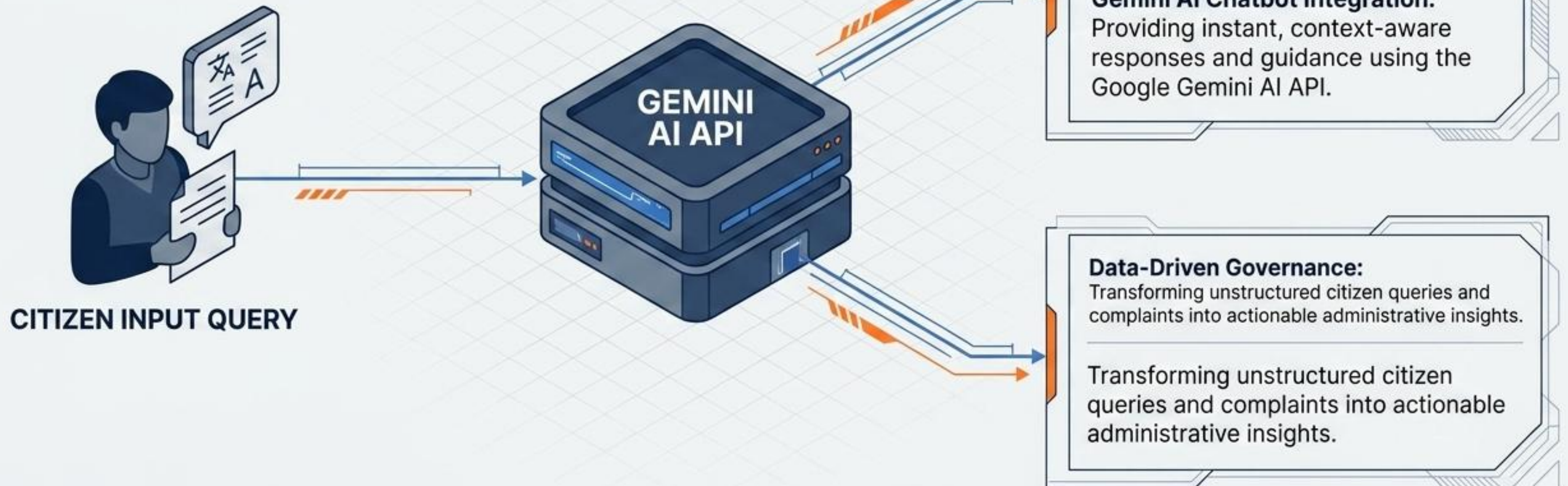
- AI-Powered Multilingual Assistance
- Real-Time Notification System
- 24/7 Smart Citizen Support

Dashboards & Control

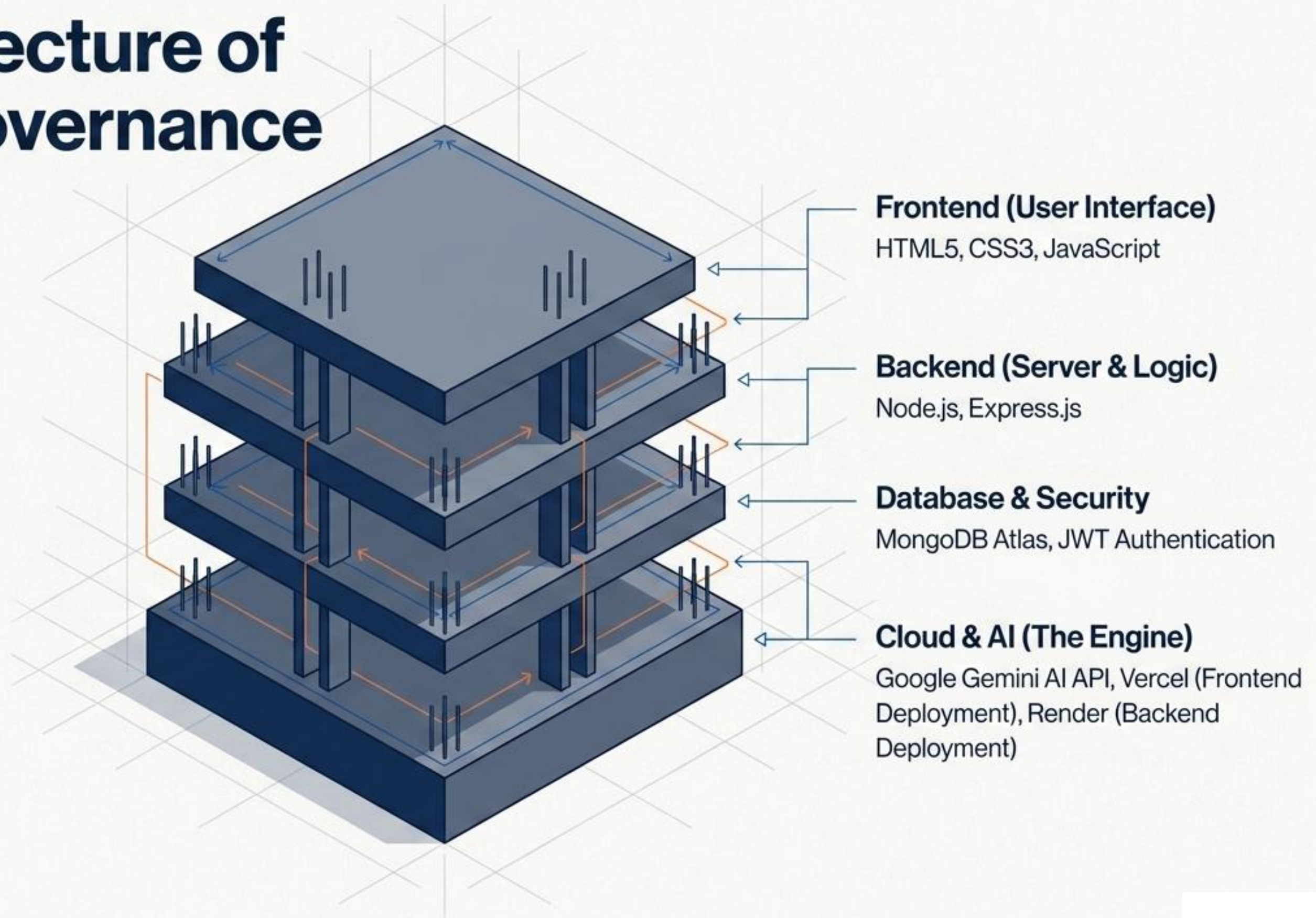
- Citizen Dashboard (Self-service tracking)
- Officer Dashboard (Task management)
- Admin/Analytics Dashboard (Reporting)

Intelligence at the Core

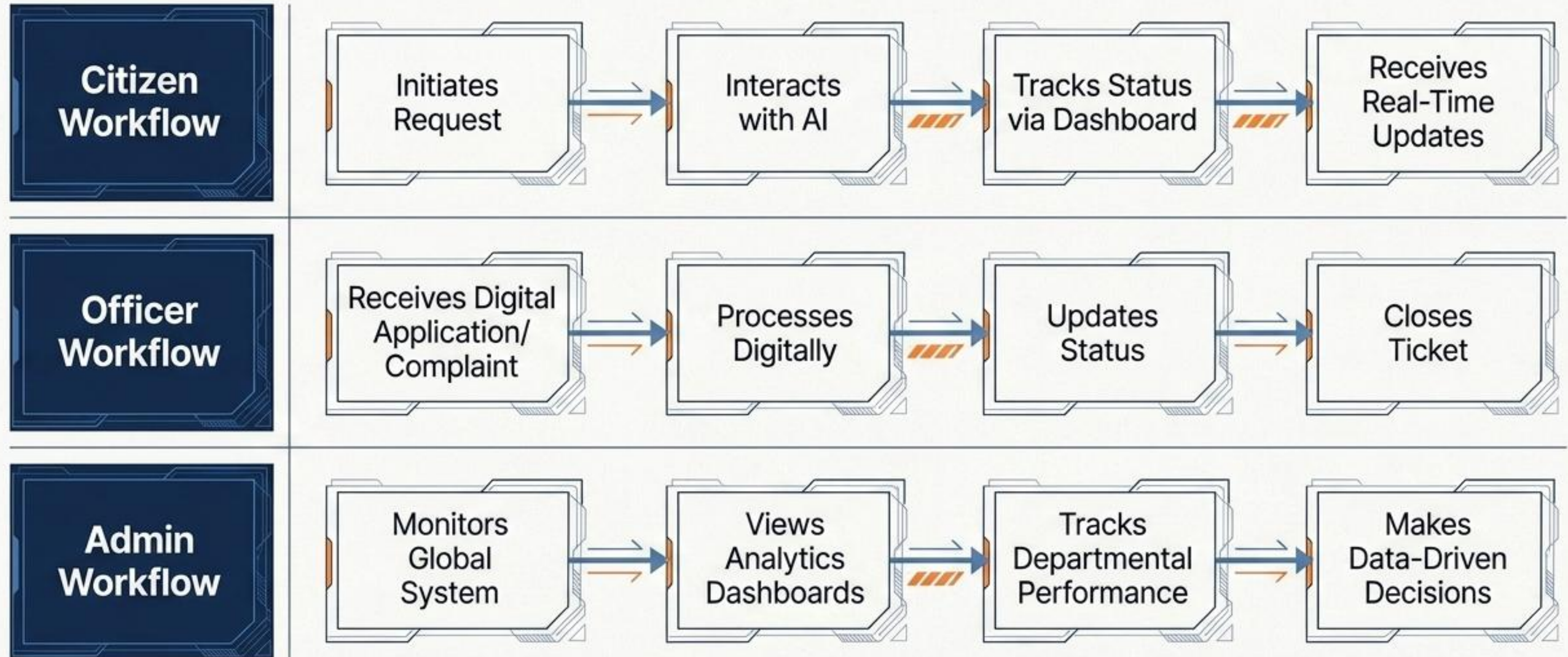
Powered by advanced LLMs, UrbanSpire AI transforms static government portals into conversational, data-driven assistants.



The Architecture of Modern Governance



Integrated Workflow Dynamics



Designing for Global Impact

UrbanSpire AI directly advances the United Nations Sustainable Development Goals:



SDG 9: Industry, Innovation, and Infrastructure

Modernizing civic infrastructure through resilient digital innovation.



SDG 11: Sustainable Cities and Communities

Creating accessible, responsive, and smart urban environments.



SDG 16: Peace, Justice, and Strong Institutions

Fostering transparency, accountability, and robust municipal institutions.

The UrbanSpire AI Impact

Radical Efficiency

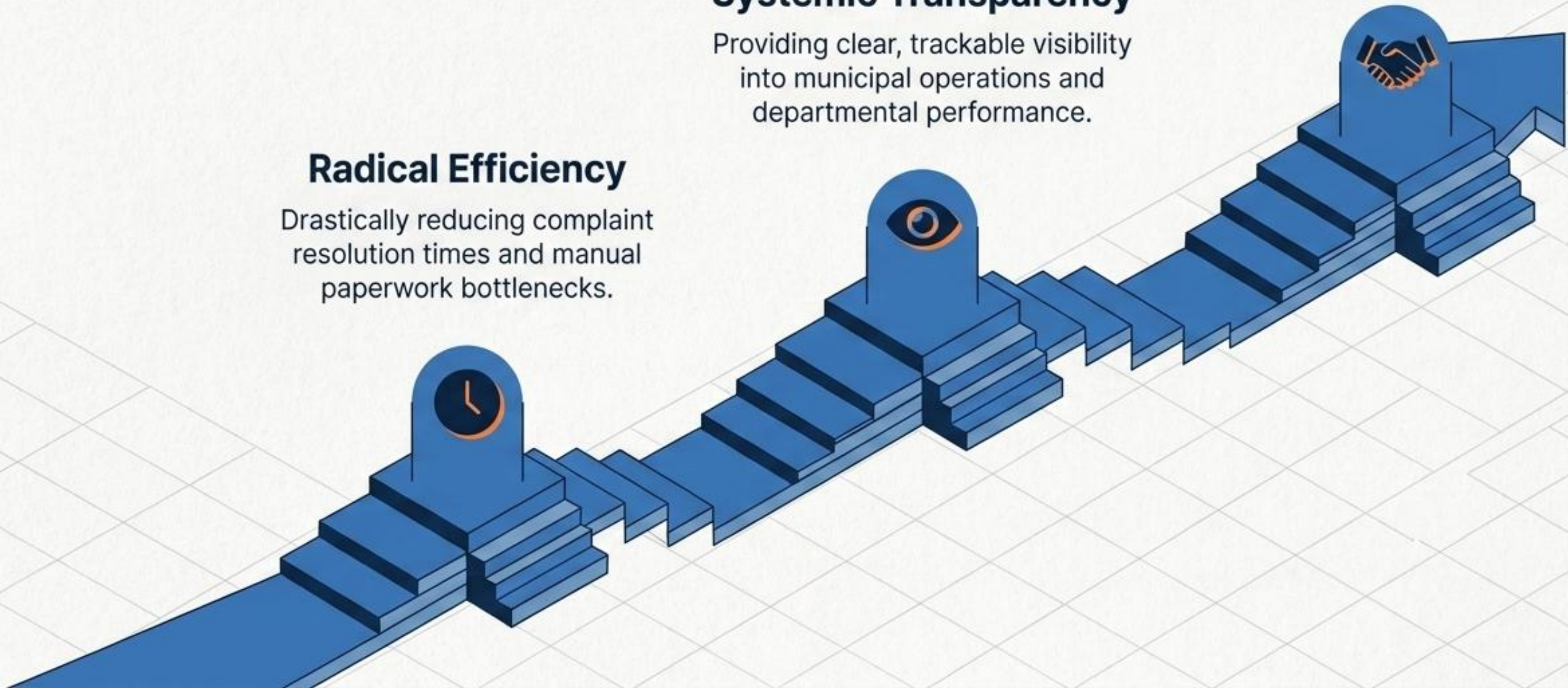
Drastically reducing complaint resolution times and manual paperwork bottlenecks.

Systemic Transparency

Providing clear, trackable visibility into municipal operations and departmental performance.

Restored Citizen Trust

Fostering deeper engagement and satisfaction through responsive, accessible, and evidence-based governance.



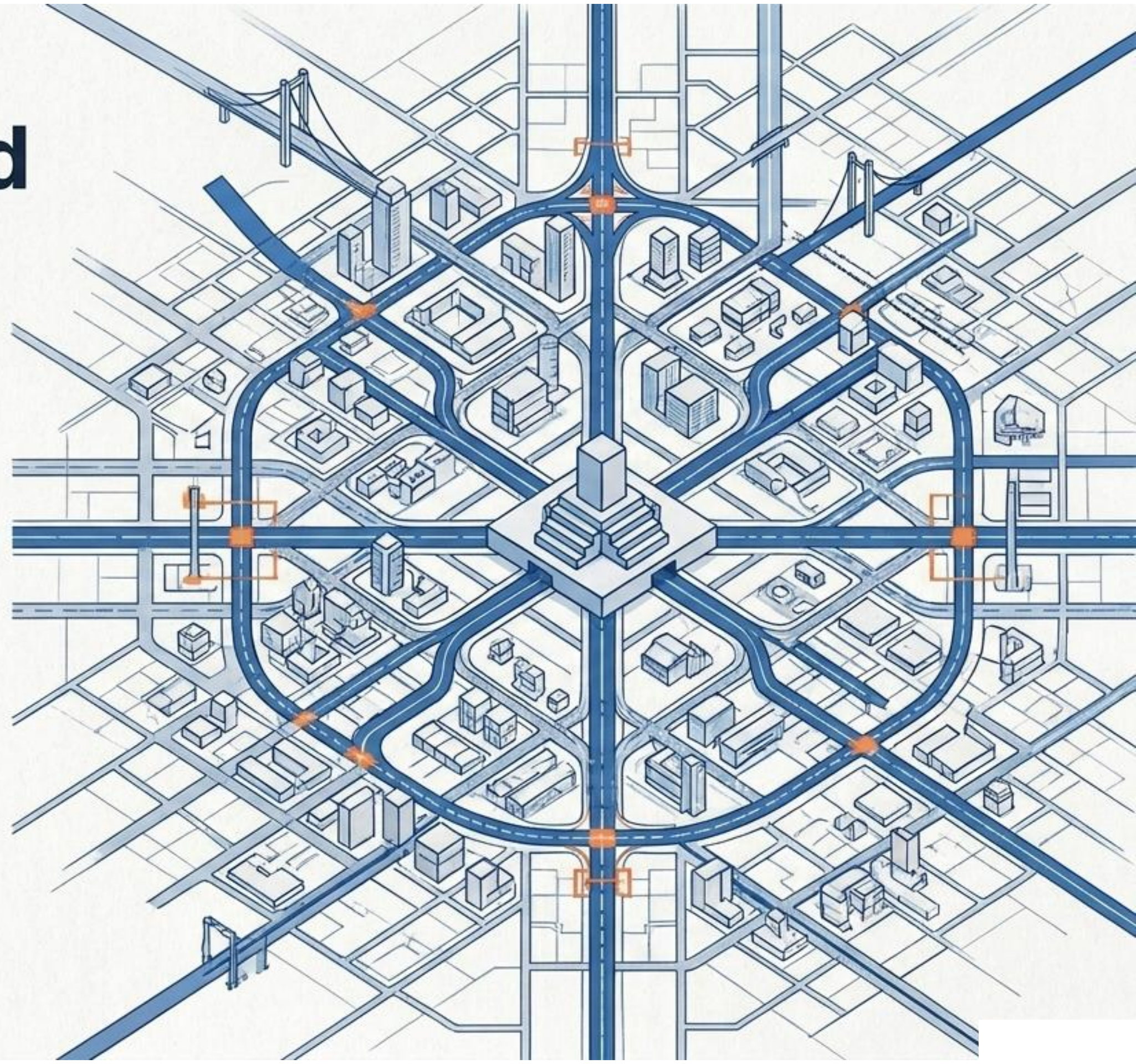
The Road Ahead

Smart City Integration

Seamlessly plugging into broader IoT frameworks, traffic grids, and automated utility systems across India's Smart Cities.

Scalable Governance

Expanding the architecture to onboard additional municipal departments, statewide deployment, and predictive urban analytics.



Project Access

GitHub Repository

<https://github.com/radhikamahajan705-afk/UrbanSpire-AI>

Live Demo (Frontend)

<https://urbanspire-ai.vercel.app/>

Thank You.